SDG BLOCKCHAIN ACCELERATOR

Technical Architecture Document

1. Project Information

- **Project Name: Green Giraffe Digital Traceability & Market Access Pilot (Kurima Programme)**
- **Challenge & UNDP Office: UNDP Angola**
- **Document Version: v1.0**

2. Overview

This project implements a blockchain-enabled agricultural traceability, quality assurance, and payment reconciliation platform built on the Cardano blockchain. The system addresses structural market access challenges faced by smallholder farmers by enabling verifiable proof of origin, quality, ESG compliance, and chain of custody from farm to buyer. By combining digital identities, crop management logs, QA verification, and escrow-based settlement, the platform ensures that only verified agricultural data and outcomes are anchored on-chain, supporting SDGs 1, 2, and 12 and enabling evidence-based rural economic transformation under the Kurima Programme.

3. System Architecture Diagram

Core Components:

- **Farmer, QA Agent, Buyer, and Admin Web Interfaces**
- **Backend Services PHP Laveral**
- **Agribalyse 3.2 LCA & ESG Engine**
- **QA & Compliance Engine**
- **Escrow & Payment Gateway (Multicaixa Express / Stanbic Bank)**
- **Cardano Blockchain (UTxO Ledger)**
- **External Data Providers (Satellite, Climate, Soil)**

Illustrative Flow (ASCII):

```
User Interface
   |
   v
Backend Services (PHP)
   |
   +--> QA & Compliance Engine
   |
   +--> LCA / ESG Engine
   |
   +--> Payment & Escrow System
   |
   v
Blockchain Adapter
   |
   v
Cardano Ledger (UTxO, Metadata, Hash Anchors)
```

External oracles (satellite, climate, deforestation) feed verified analytics into off-chain services whose outputs are cryptographically anchored on-chain.

4. Blockchain Design

Smart Contracts / On-Chain Logic

The system uses Cardano primarily as a **trust anchor**, not a computation-heavy execution layer.

- **Purpose:**
  - Anchor immutable records for batches, QA verification, compliance summaries, lab results, and reconciliation reports.
- **On-chain Objects:**
  - Hash of structured JSON documents
  - Metadata references (Farmer ID, Batch ID, QA ID, timestamps)
- **Validator Scripts:**

    - Access control validation
    - Authorized role signing (QA, Admin)
    - Prevention of overwriting (append-only design)

Datum Structure

- Entity ID (batch, farmer, report)
- Document hash
- Version number
- Timestamp
- Role signature reference

Redeemer Structure

- Create (new record)
- Append (new version)
- Lock (final inspection completion)

UTxO Model Usage

- Each verified artifact generates a new UTxO reference
- No state mutation; corrections create new outputs
- Downstream processes reference latest valid UTxO

Security Considerations

- Role-based signing (QA vs Admin separation)
- Immutable append-only records

- On-chain proof prevents tampering and retroactive edits
- Financial data excluded from QA role access

5. Data Flow & Transaction Lifecycle

Example:

1. Farmer submits production data (off-chain).
2. QA Agent verifies data via structured forms.
3. Backend generates structured JSON + cryptographic hash.
4. Hash and metadata anchored to Cardano.
5. Off-chain systems update dashboards and compliance status.
6. Upon purchase, Final Inspection references existing on-chain records only.
7. Reconciliation results are anchored and trigger escrow settlement logic.

6. Off-chain Components

Backend Services:
PHP laravel microservices handling identity, QA workflows, ESG/LCA calculations.

Blockchain Integration:
Cardano node access via APIs; transaction submission and monitoring.

External Data Providers:

- NASA HLS (crop & land monitoring)
- Prithvi (analysis layer)
- Global Forest Watch (deforestation risk)

- OpenWeatherMap Agro API
- iSDAsoil API

Dashboards:

Role-based dashboards for farmers, QA agents, buyers, and UNDP/Admin stakeholders.

7. Sandbox/Testnet Results

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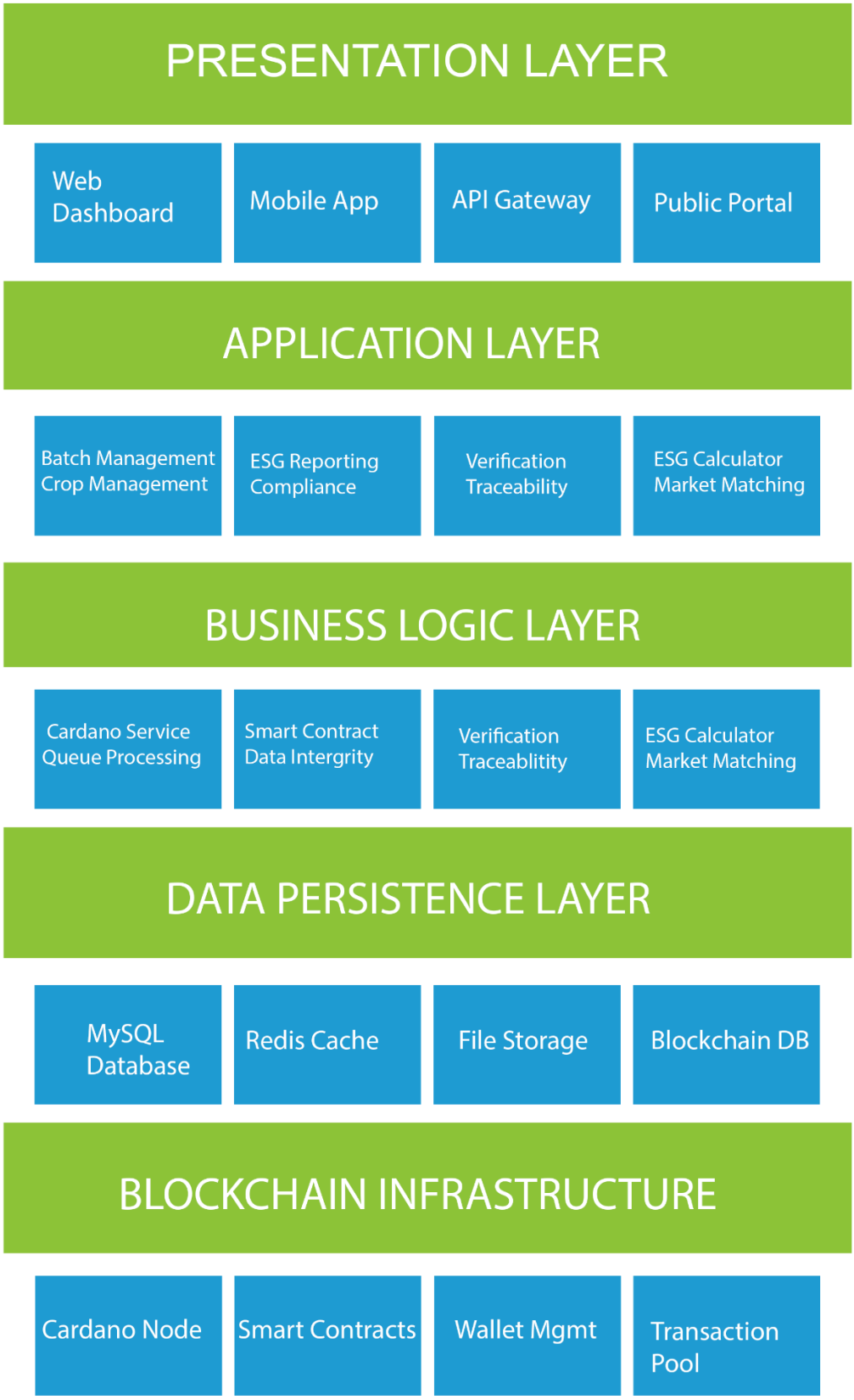
8. Tools and Environments Used

- **Blockchain:** Cardano (Preprod / Testnet)
- **Node & APIs:** cardano-node, Cardano transaction APIs
- **Languages:** PHP, Laravel (frontend)

- **Cloud Hosting:** Wasabi, Railway
- **Payments:** Multicaixa Express, Stanbic Bank
- **Data & Analytics:** Agribalyse 3.2 LCA, satellite & climate APIs

9. Remaining Considerations / Next Steps

- Expand on-chain metadata standards for cross-border interoperability
- Conduct third-party security review of signing & role separation
- Scale testing under higher transaction volumes
- Integrate automated reporting for UNDP M&E requirements
- Prepare production deployment for Kurima pilot provinces


PRESENTATION LAYER
Web Dashboard
Mobile App
API Gateway
Public Portal
APPLICATION LAYER
Batch Management Crop Management
ESG Reporting Compliance
Verification Traceability
ESG Calculator Market Matching
BUSINESS LOGIC LAYER
Cardano Service Queue Processing
Smart Contract Data Intergrity
Verification Traceablitity
ESG Calculator Market Matching
DATA PERSISTENCE LAYER
MySQL Database
Redis Cache
File Storage
Blockchain DB
BLOCKCHAIN INFRASTRUCTURE
Cardano Node
Smart Contracts
Wallet Mgmt
Transaction Pool